

CLASS 10 SCIENCE

Chapter 1: Chemical Reactions and Equations

PYQ Question Bank | Easy · Medium · Hard | 42 Questions

17 Questions
EASY

14 Questions
MEDIUM

11 Questions
HARD

Total 62 Marks
WEIGHTAGE

SECTION A — EASY (1 Mark & 2 Mark)

1M

Q1. Define a chemical reaction.

[1 Mark] Source: CBSE 2018

1M

Q2. What happens to the mass of reactants and products in a chemical reaction?

[1 Mark] Source: CBSE 2019

1M

Q3. Name the type of reaction that takes place when magnesium ribbon burns in air.

[1 Mark] Source: CBSE 2020

1M

Q4. What is a balanced chemical equation? Why should a chemical equation be balanced?

[1 Mark] Source: CBSE 2017

1M

Q5. What are the physical states of reactants and products represented by the symbols (s), (l), (g) and (aq) in a chemical equation?

[1 Mark] Source: CBSE 2019

1M

Q6. What is a combination reaction? Give one example.

[1 Mark] Source: CBSE 2020

1M

Q7. What is a decomposition reaction? Give one example with a balanced equation.

[1 Mark] Source: CBSE 2021

1M

Q8. What is an exothermic reaction? Give one example.

[1 Mark] Source: CBSE 2018

1M

Q9. What is an endothermic reaction? Give one example.

[1 Mark] Source: CBSE 2019

1M **Q10.** Define oxidation and reduction in terms of gain or loss of oxygen.

[1 Mark] Source: CBSE 2020

1M **Q11.** What is rancidity? How can it be prevented?

[1 Mark] Source: CBSE 2022

1M **Q12.** What is corrosion? Give one example of a metal that undergoes corrosion.

[1 Mark] Source: CBSE 2021

1M **Q13.** Identify the type of reaction: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

[1 Mark] Source: CBSE 2018

1M **Q14.** Name the reducing agent in the following reaction: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

[1 Mark] Source: CBSE 2020

2M **Q15.** Write the balanced chemical equation for the reaction between iron and steam.

[2 Marks] Source: CBSE 2019

2M **Q16.** Why is the addition of hydrogen called a reduction reaction? Give one example with a chemical equation.

[2 Marks] Source: CBSE 2022

1M **Q17.** What is a displacement reaction? Give one example.

[1 Mark] Source: CBSE 2017

SECTION B — MEDIUM (2 Mark & 3 Mark)

2M **Q18.** Balance the following chemical equation and identify the type of reaction: $\text{Fe}_2\text{O}_3 + \text{Al} \rightarrow \text{Al}_2\text{O}_3 + \text{Fe}$

[2 Marks] Source: CBSE 2019

2M **Q19.** What is a double displacement reaction? How does it differ from a displacement reaction? Give one example of each.

[2 Marks] Source: CBSE 2020

2M **Q20.** Write the balanced chemical equation for the decomposition of lead nitrate on heating. What type of decomposition is this?

[2 Marks] Source: CBSE 2018

3M **Q21.** What is a precipitation reaction? Explain with the help of an example. Write the balanced chemical equation.

[3 Marks] Source: CBSE 2021

3M **Q22.** Explain thermite reaction with a balanced chemical equation. What type of reaction is it and why?

[3 Marks] Source: CBSE 2020

2M **Q23.** The colour of copper sulphate solution changes when iron nails are dipped into it. Explain with a balanced chemical equation.

[2 Marks] Source: CBSE 2019

3M **Q24.** Identify the oxidising agent and reducing agent in the following reactions: (a) $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ (b) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$

[3 Marks] Source: CBSE 2022

2M **Q25.** Why do we apply paint on iron articles? What kind of chemical change are we trying to prevent?

[2 Marks] Source: CBSE 2018

3M **Q26.** What happens when silver chloride is kept in sunlight? Write the balanced equation and name the type of reaction.

[3 Marks] Source: CBSE 2021

2M **Q27.** Why are chips manufacturers usually flush bags of chips with gas like nitrogen? Explain in terms of oxidation.

[2 Marks] Source: CBSE 2020

3M **Q28.** Write balanced chemical equations for the following: (a) Burning of natural gas (methane) in air (b) Electrolysis of water (c) Reaction of zinc with dilute sulphuric acid

[3 Marks] Source: CBSE 2019

2M **Q29.** Define the law of conservation of mass. Verify it using the reaction between sodium sulphate and barium chloride solutions.

[2 Marks] Source: CBSE 2018

2M **Q30.** What is observed when a few drops of dilute HCl are added to silver nitrate solution? Write the balanced equation.

[2 Marks] Source: CBSE 2022

3M

Q31. Explain with examples: (a) Thermal decomposition (b) Electrolytic decomposition (c) Photolytic decomposition

[3 Marks] Source: CBSE 2021

SECTION C — HARD (3 Mark & 5 Mark)

3M

Q32. How do you distinguish between a physical change and a chemical change? Give two examples of each. Identify which of the following is a chemical change: melting of wax, souring of milk, dissolving salt in water, rusting of iron.

[3 Marks] Source: CBSE 2020

5M

Q33. What is redox reaction? Explain oxidation and reduction in terms of electron transfer with suitable examples. Identify the oxidising and reducing agents in the reaction: $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$

[5 Marks] Source: CBSE 2019

5M

Q34. Write the steps you would use to balance a chemical equation. Balance the following equation step by step: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$ Also name the type of reaction and identify oxidising/reducing agents.

[5 Marks] Source: CBSE 2021

3M

Q35. Why should a magnesium ribbon be cleaned before burning in air? Write the balanced chemical equation for the combustion of magnesium in air. What type of oxides does magnesium form?

[3 Marks] Source: CBSE 2018

5M

Q36. Describe any three methods of preventing corrosion of metals. Explain with chemical equations where applicable. Why is iron more prone to corrosion than gold?

[5 Marks] Source: CBSE 2022

3M

Q37. State the activity series of metals. Using it, predict whether the following reactions will occur or not. Give reasons: (a) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ (b) $\text{Cu} + \text{FeSO}_4 \rightarrow \text{CuSO}_4 + \text{Fe}$

[3 Marks] Source: CBSE 2020

5M

Q38. Explain the following observations with balanced chemical equations: (a) A white precipitate is formed when sodium sulphate solution is mixed with barium chloride solution. (b) Lime water turns milky when CO_2 is passed through it, but clears up on excess CO_2 . (c) A silver article turns black when exposed to air for long.

[5 Marks] Source: CBSE 2021

3M

Q39. What is the role of a catalyst in a chemical reaction? Explain the decomposition of hydrogen peroxide in the presence of manganese dioxide with a balanced equation. How does a catalyst affect the rate of reaction?

[3 Marks] Source: CBSE 2019

5M

Q40. Distinguish between combination, decomposition, and displacement reactions. Give one balanced equation for each. Also identify which of these reactions are exothermic and which are endothermic, with justification.

[5 Marks] Source: CBSE 2022

3M

Q41. A student mistakenly adds excess CO_2 to lime water and notices the milky precipitate dissolves. Write balanced equations for both steps and explain what type of reactions are occurring.

[3 Marks] Source: CBSE 2020 (Application)

5M

Q42. Analyse the following statement: 'All combustion reactions are combination reactions but all combination reactions are not combustion reactions.' Justify with two examples each and relevant balanced equations.

[5 Marks] Source: CBSE 2022 (HOT)